SMU
SINGAPORE MANAGEMENT UNIVERSITY

School of
Information Systems

SE201 | AI-Driven Software Engineering

Project Briefing

SMU Classification: Restricted

Overview

- Design, build and evaluate a working AI-enabled solution
  - Large language model at its core
- Choose one of two routes:
  - Improve an existing product or practice
  - Prototype a new use case made possible by LLMs
- Apply techniques such as RAG, agentic workflows, or tool integration
- A successful project shows:
  - Clear architecture and data flow
  - Rational use of AI to solve the problem
  - Clear metrics for measuring success
- A command-line interface and a locally run model are acceptable
- Throughout the project
  - Disclose how you use AI
  - Assess your solution's limitations, risks, and ethical implications

Assessments and dates

Component	Due date	Total	Team	Individual
Proposal	2 October (Friday) at 23:59	5%	5%	
Checkpoint	18 October (Sunday) at 23:59	5%	5%	
Presentation / Deliverables	10 November (Tuesday) at 12:00	20%	15%	5%
Peer Evaluation	17 November (Tuesday) at 23:59	5%		5%
Total		35%	25%	10%

- The instructor assesses the presentation and deliverables
  - See rubrics in Appendix A of the project specification
- Your teammates assess you via peer evaluation
- Submit everything through eLearn / Assignments

Proposal (5%)

- Submit a two-page document
  - **Problem statement**
    - The problem or new use case
    - The intended users or stakeholders, and why the problem matters to them
  - **Justification for using AI (LLM)**
    - Why AI is a suitable solution
    - Describe what AI offers that a conventional, non-AI alternative does not
  - **Proposed solution**
    - Main features, with a high-level architecture diagram
      - Main components and how data flows between them
    - Techniques you will use and why you chose them
    - Models, APIs, frameworks, and data sources you will use
      - Confirmation that you can access them!

Proposal (5%)

- Proposal document (continued)
  - **Evaluation plan**
    - Metrics, test cases, or baseline
    - Your test data or scenarios
    - What counts as success?
  - **Checkpoint targets**
    - Deliverables you commit to showing at the checkpoint
    - At a minimum
      - One working core feature
      - Main architecture components in place and connected, even if some are stubbed

Checkpoint (5%)

- This is a progress check. You are not expected to have met every goal, and problems you find are useful information, so document them
- Your submission must include:
  - Read access to a shared repository with a README containing setup instructions, and commits from every team member
  - Main architecture components in place and connected, with any stubbed components documented
  - At least one core feature demonstrated, as described in your proposal
  - Your chosen AI technique working in the prototype, even in a limited form
- It must also include a document containing:
  - Results from at least one round of testing, including any failures you observed
  - Your progress against the checkpoint targets from your proposal
  - Any changes to scope, design, or technology since the proposal, with your reasons
  - Your AI use disclosure log to date

Final presentation and Deliverables (20%)

- Presentations take place in class in Week 13
  - 20 minutes per team, made up of a 15-minute presentation and 5 minutes of questions
- All members must present and be able to answer questions
- Phones and on-screen notes are not allowed, but written or printed notes are.
- Your presentation should cover:
  - The problem and your justification for using AI
  - Your technical design and choice of model
  - Evaluation
  - Responsible AI use
  - Your conclusions
- The rubric has two parts:
  - **Team (15%):** problem and use of AI, design and implementation, evaluation, responsible AI use, and team collaboration
  - **Individual (5%):** presentation skills and answers to questions

Deliverables

- Submit a zipped file by 10 November at 12:00 (noon)
- The zipped file must contain:
  - Your slides
  - A repository link or snapshot
    - README containing setup instructions and runnable examples
  - Supporting documents such as test results, run logs, or material cut from the presentation
  - Your AI use disclosure log
- **There is no report.** Your slides must be readable on their own or supported by speaker notes.

Peer evaluation (5%)

- EVERY student must rate the others in their team
- Take the peer evaluation seriously
- There is a penalty if you do not complete it
- Scores run from 1 to 10 in five categories
  - Team player
  - Communication skill
  - Quality of work
  - Commitment
  - Contribution
- Be frank, and do not give everyone the same score
- **Raise major team issues with your instructor**
  - Do not wait for the peer evaluation

Disclosure of AI tool use

- You may use AI tools during development
- Your team must record how it uses them in an AI use disclosure log
  - A separate document will describe the log
- You submit the log at the checkpoint and again with your final deliverables
- Track *how* you use the tools
  - What did you learn about the best way to use the tool?
  - What are the tool's strengths and weaknesses?
  - What hard lessons have you learned?